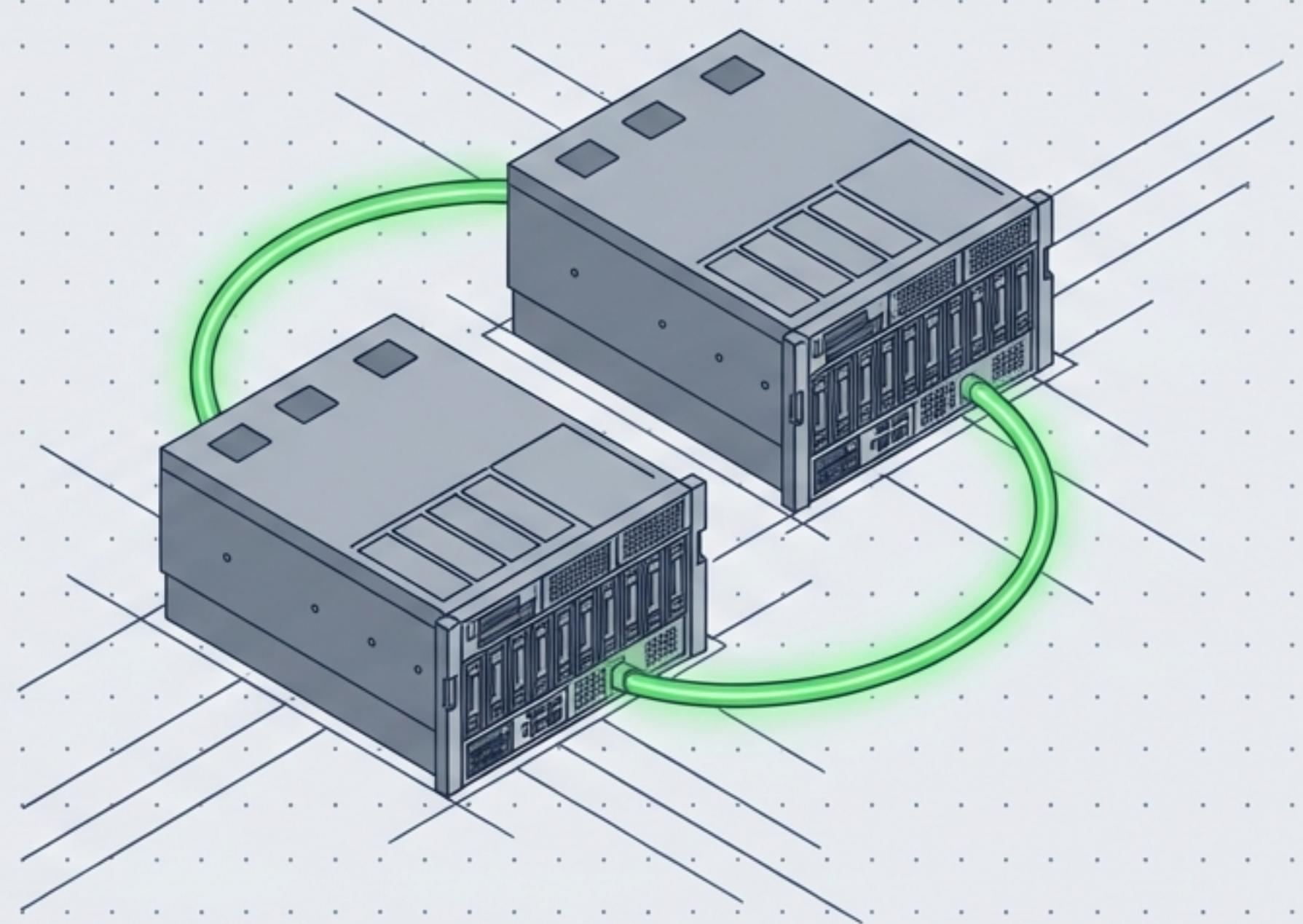


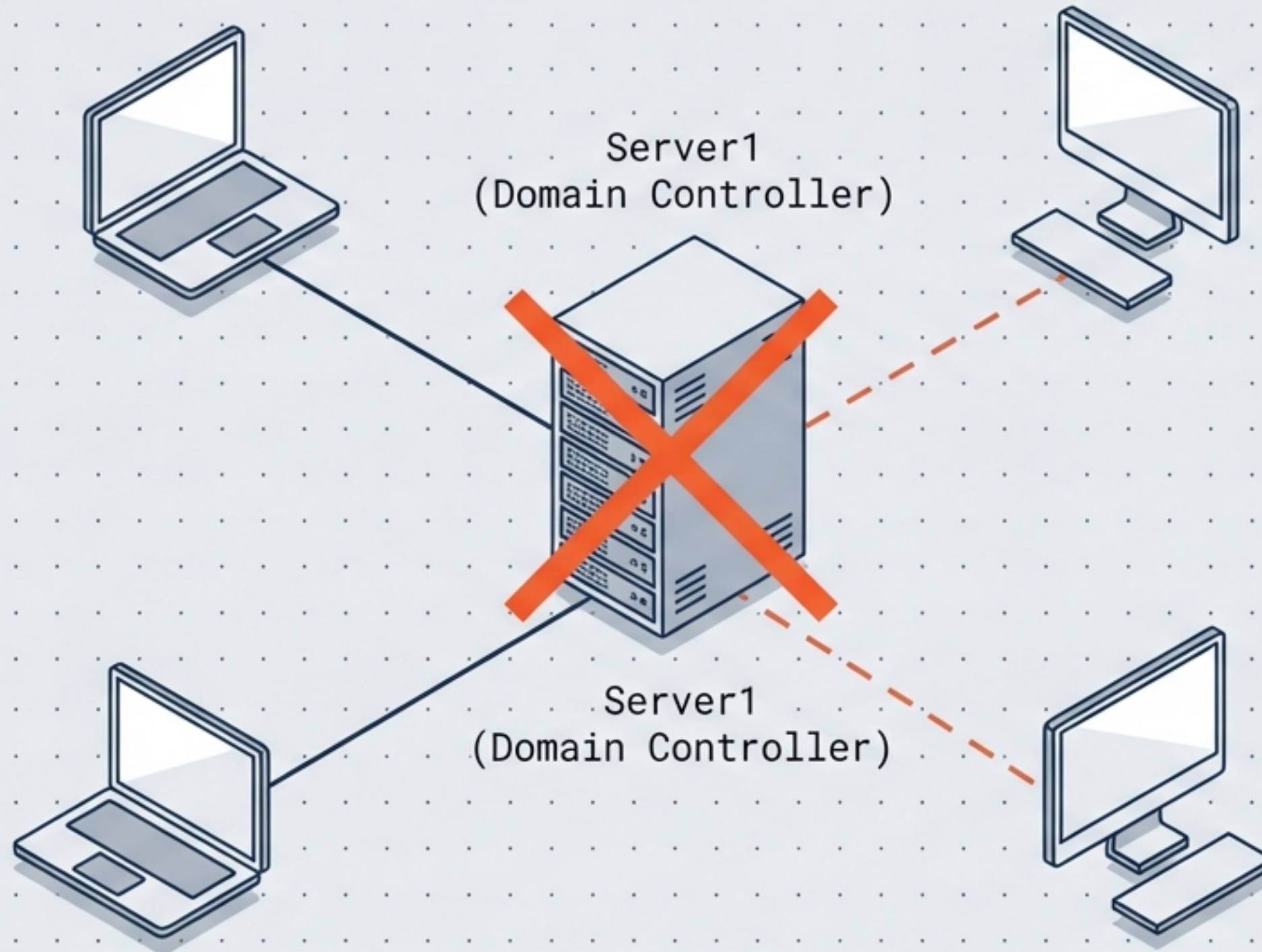
Architecting High Availability:

Deploying an Additional Domain Controller

- **Primary Focus:** Windows Server 2019 Domain Expansion
- **Core Objectives:** Fault Tolerance, Load Balancing, Service Continuity
- **Target Domain:** hp.com



The Vulnerability of a Single Point of Failure (SPOF)



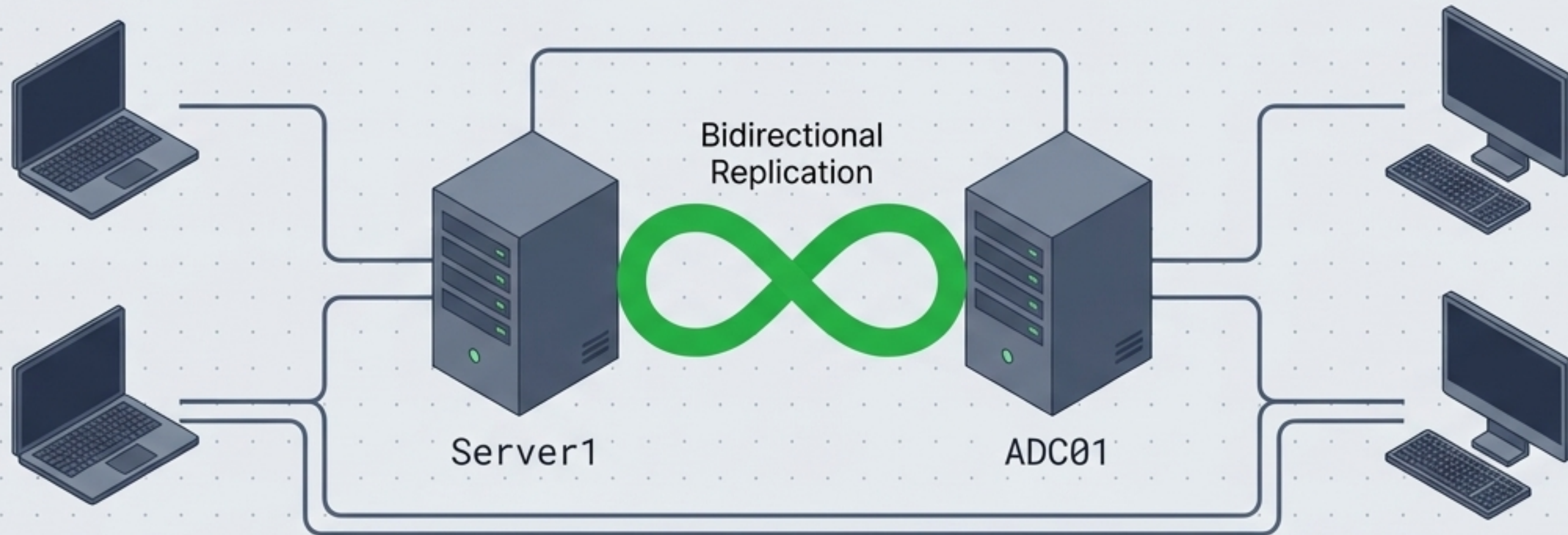
The Risk: A single server houses the entire Active Directory database.

The Impact: If the server falls offline, the network halts.

Consequences:

- Users cannot authenticate or log in.
- Network resources become unreachable.
- Complete loss of Active Directory services.

Securing the Network with an Additional Domain Controller



Database Replication

A live copy of the Active Directory database is continuously synced to the new ADC.

Load Balancing

Authentication requests are distributed across both servers, optimizing network traffic.

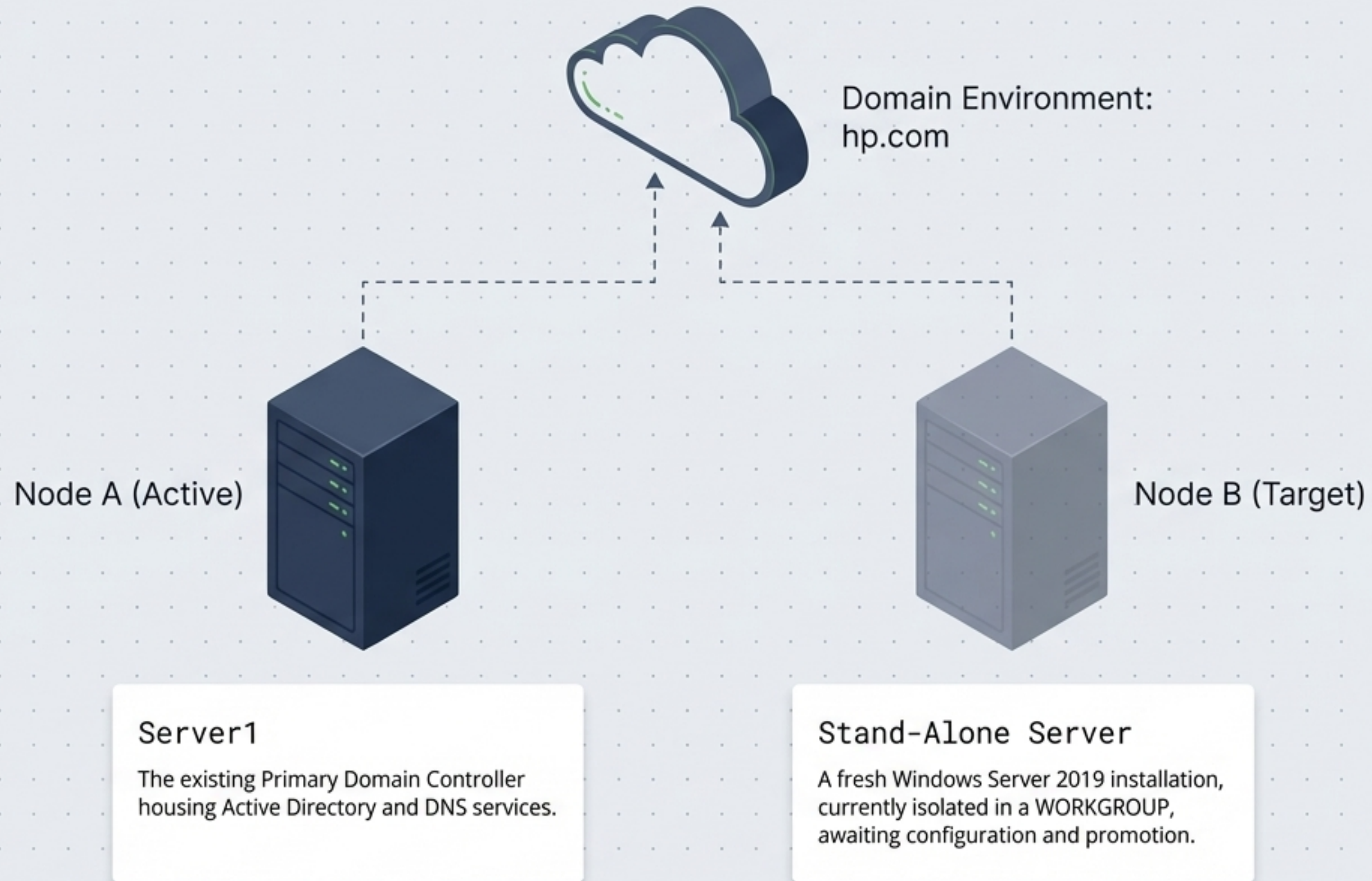
High Availability

If one server fails, clients seamlessly authenticate against the active secondary node.

Architectural Diagnostics: Single vs. Dual Domain Controllers

	Single DC (SPOF)	Dual DC (High Availability)
User Login (During Outage)	Impossible	Continuous & Seamless
Database Security	Vulnerable to hardware failure	Redundant & Replicated
Traffic Load	Bottlenecked at one node	Distributed across nodes
Maintenance Windows	Requires total network downtime	Zero-downtime rolling updates

The Pre-Flight Network Topology

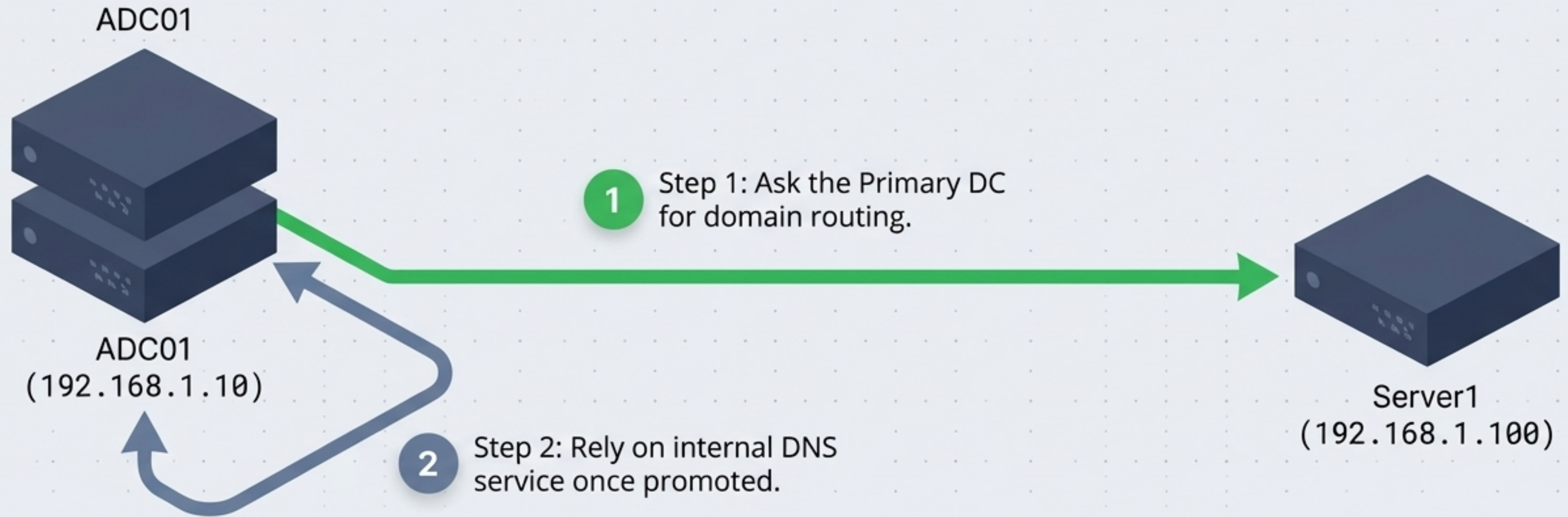


Node Configuration Parameters

Primary Node	
Hostname:	Server1
IPv4 Address:	192.168.1.100
Preferred DNS:	192.168.1.100 (Self)
Initial Status:	Domain Controller

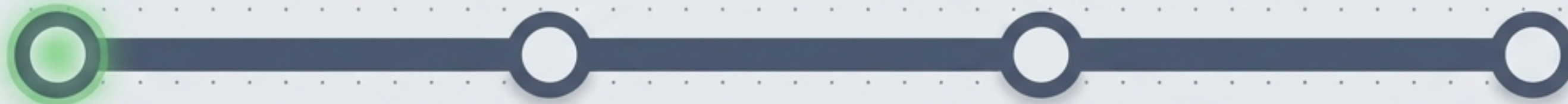
Target Node	
Hostname:	ADC01
IPv4 Address:	192.168.1.10
Preferred DNS:	192.168.1.100 (Points to Server1)
Alternate DNS:	192.168.1.10 (Self)
Initial Status:	Workgroup

The Cross-DNS Pointer Model



The Logic: Before promotion, ADC01 must find the hp.com domain. It relies on Server1's DNS to map the network. Post-promotion, ADC01 will host its own DNS service, requiring a loopback address as a fallback.

The Four Phases of ADC Deployment



Phase 1: Network Prep

Assign manual IPv4/DNS and verify connectivity.

Phase 2: Trust Establishment

Join the domain and authenticate.

Phase 3: Role Injection

Install Active Directory Domain Services (AD DS).

Phase 4: Node Promotion

Execute the Domain Controller promotion engine.

Phase 1: Network Preparation & Diagnostics

The Interface

UI Properties

IP address:

Preferred DNS server:

Alternate DNS server:

Static Addressing: Assign the manual IPv4 configuration mapped in the Pre-Flight Blueprint.

The Terminal

```
> ping 192.168.1.100 ✓  
  (Validates network reachability to Server1)  
  
> nslookup server1.hp.com ✓  
  (Validates DNS resolution is functioning properly)
```


Phase 2: Establishing Domain Trust



Target Domain

Change the computer's membership from Workgroup to Domain, targeting hp.com.

Authentication

Joining a domain requires high-level clearance. Supply the Primary DC's administrator credentials (hpit\administrator) to authorize the handshake.

Reboot Required

The system must restart to initialize the new domain membership policies.

Phase 3: Injecting the AD DS Role

Add Roles and Features

Status

- > ☒ File and Storage Services
- ☐ Print and Document Services
- ☐ Power iPads
- ☐ Active Client
- ✓ ☒ Active Directory Domain Services
 - ✓ ☒ Add Features that are required for AD DS
- ☐ ADP Features
- ☐ Furuntime Management
- ☐ Properiy
- ☐ Postarlogment Services
- ☐ Print & Ciromcy Services
- ☐ Server Space

Role-Based Installation

Select the target server (ADC01) from the server pool.

Core Binaries

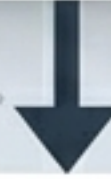
Check Active Directory Domain Services. This installs the foundational framework, management tools, and dependencies required to house the database.

Note:

Installing the role does not create a DC; it merely prepares the server for promotion.

Phase 4: The Promotion Engine

Click “Promote this server to a domain controller”



Select Deployment Operation -> “Add a domain controller to an existing domain” (hp.com)



Define DSRM Password

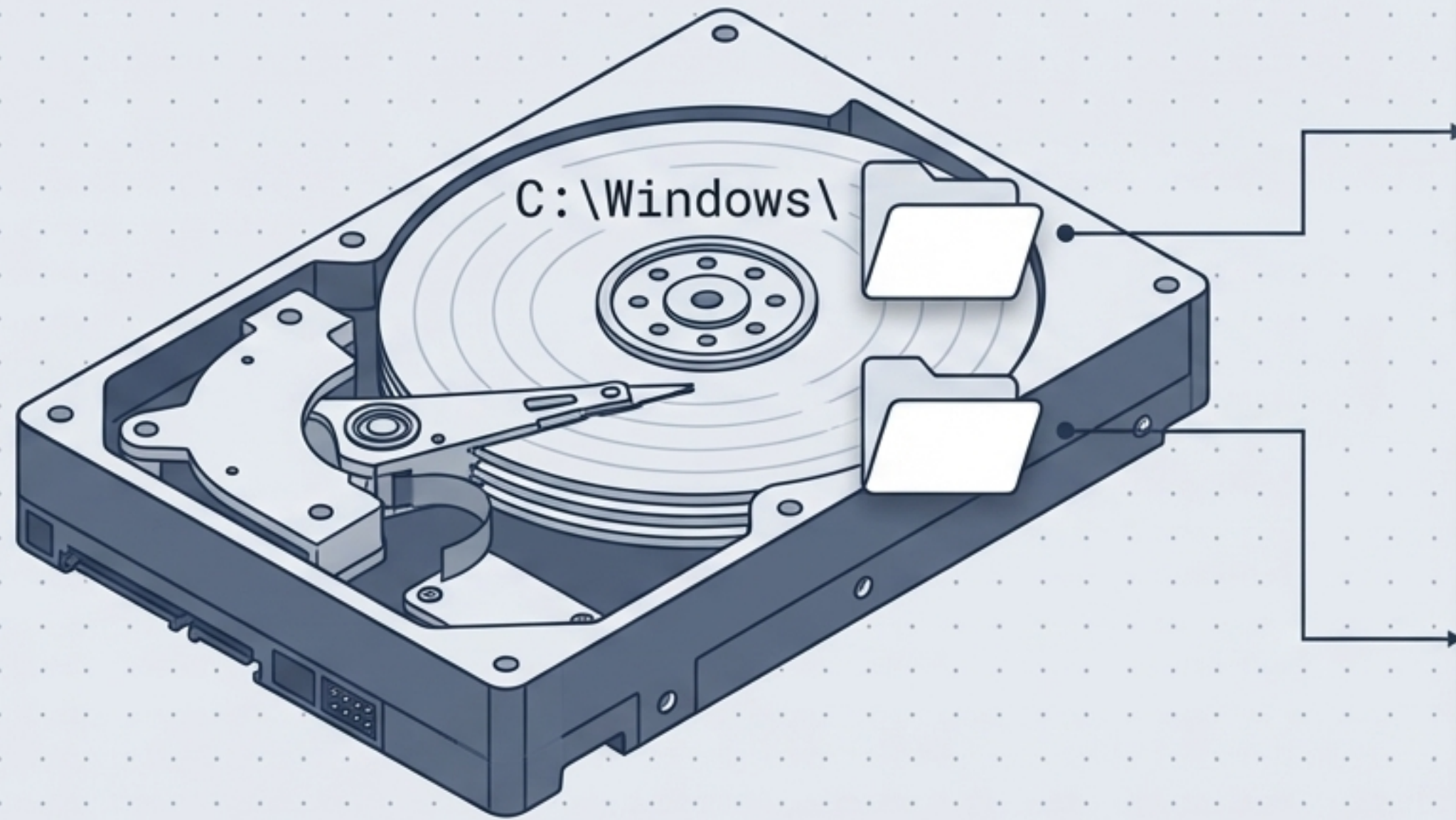
Targeting the Environment:

Ensure the deployment specifically targets an existing domain, not a new forest.

DSRM Configuration:

Set the Directory Services Restore Mode password. This is a critical fail-safe credential required if the Active Directory database ever needs to be manually restored from a backup.

Underlying Storage Architecture



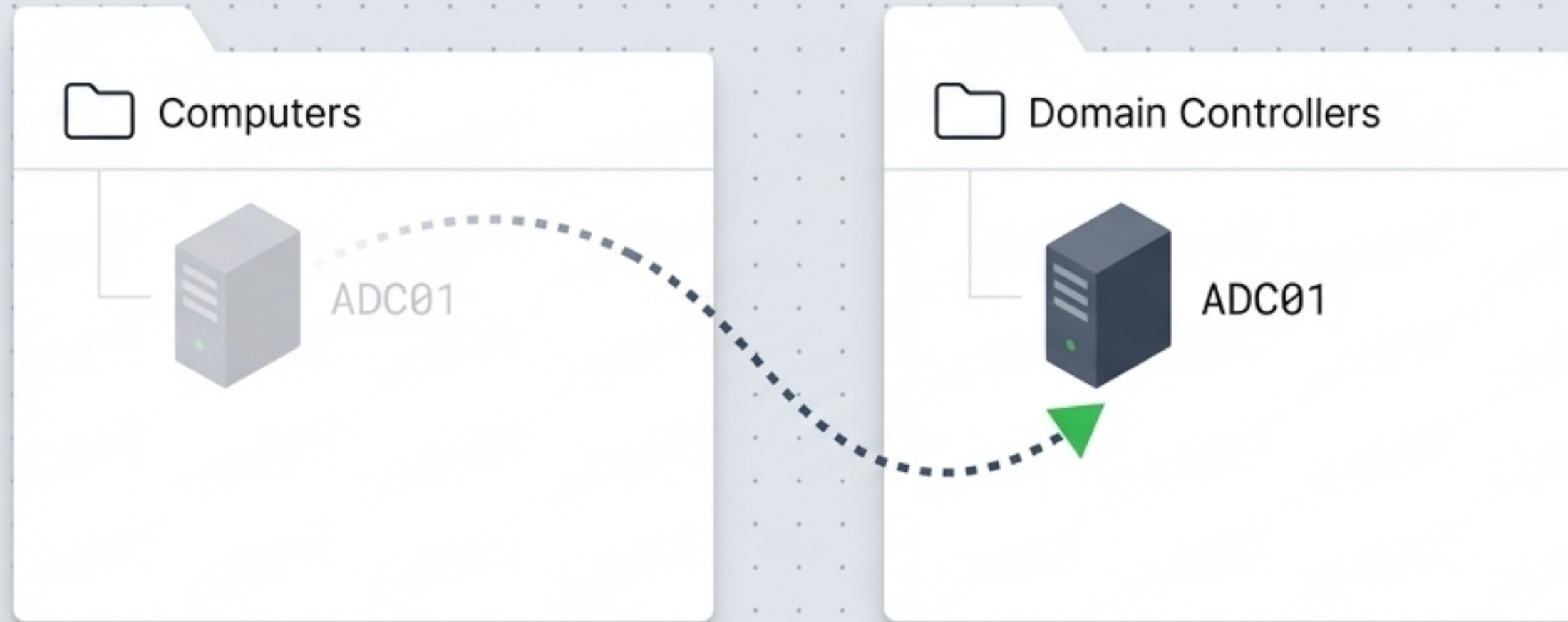
Database & Logs: C:\Windows\NTDS

The physical location of the Active Directory database file (ntds.dit) and transaction logs.

Group Policy & Scripts: C:\Windows\SYSVOL

The System Volume folder, which automatically replicates group policies, logon scripts, and security information across all Domain Controllers.

The Topology Shift: Validating the Promotion



Pre-Promotion

ADC01 existed merely as a standard domain computer.

Post-Promotion

Upon restart, ADC01 is elevated. Verifying its presence in the Domain Controllers OU on Server1 confirms the promotion engine executed successfully.

Proof of Concept: The Replication Loop



The Final Test

Create a test user explicitly on the new ADC01 server.

The Result

Without manual intervention, the user object instantly replicates to the primary Server1.

Synthesis

Load balancing and high availability are now active. Changes made on any node map synchronously across the entire domain architecture. Network failure at a single point is no longer fatal.